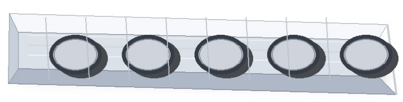
PRODUCT	MEMBER #	MIL	GAGE	COATING
Smart Stud	600S125-18	18	25	G40

600S125-18 Product Submittal

PHYSICAL PROPERTIES

Web Depth	6"
Flange	1.25"
Return (lip)	0.188"
Approx. Weight (lb/ft)	0.5535
Area (in^2)	0.1624
Design Thickness	0.0188
Minimum Thickness	0.0179
Yield Strength	33 KSI
Coating	G40

PHOTO-STYLE PRODUCT VIEW



Illustrative view - not a manufacturer photograph

GROSS PROPERTIES

Gross section moment of inertia about X-X axis (Ix)	0.7786
Section modulus about X-X axis (Sx)	0.2595
Radius of gyration about X-X axis (Rx)	2.189
Gross section moment of inertia about Y-Y axis (Iy)	0.0237
Radius of gyration about Y-Y axis (Ry)	0.382

EFFECTIVE / NET PROPERTIES

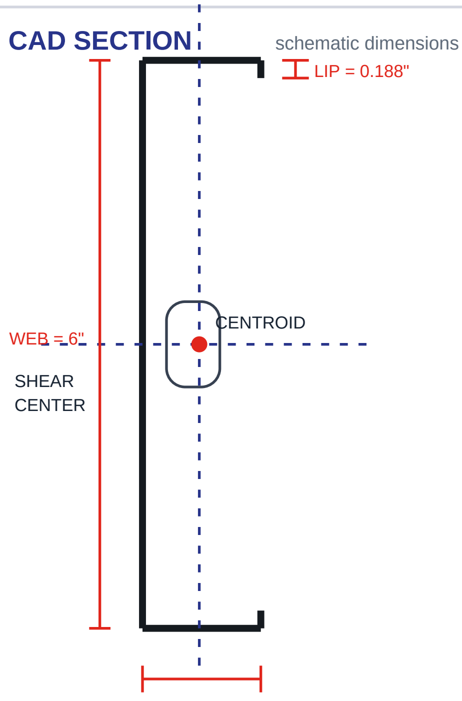
Effective moment of inertia X-axis (Ixe)	Source-limited
Effective section modulus X-axis (Sxe)	Source-limited
Allowable moment (Mal)	Source-limited
Allowable shear (Vay)	Source-limited

TORSIONAL PROPERTIES

St. Venant torsional constant (Jx1000)	0.0191
Torsional warping constant (Cw)	0.1715
Polar radius of gyration (Ro)	2.308

CAD SECTION

schematic dimensions



Mil 18 / Fy 33 ksi

NOTES

Code Approvals & Performance Standards

- AISI S100-16 (2020) w/S2-20 North American Specification for the Design of Cold-Formed Steel Structural Members.
- AISI S240-20 North American Standard for Cold-Formed Steel Structural Framing.
  - Compliant to ASTM C955; IBC replaced ASTM C955 with AISI S200 in IBC 2015 and AISI S240 in IBC 2018.
  - Section A3 Material - Chemical and mechanical requirements (referencing ASTM A1003/A1003M).
  - Section A4 Corrosion Protection (referencing ASTM A653/A653M).
  - Section A5 Products - Thickness, shapes, tolerances, identification.
  - Section C Installation (referencing ASTM C1007).
- AISI S202-20 Code of Standard Practice for Cold-Formed Steel Structural Framing.
  - Section F3 Delivery, Handling, and Storage of Materials.
- IBC 2024 International Building Code.
- ICC-ES ESR-5837 Structural Studs and Track.
  - ESR-5837 Catalog Refined Metal Technical Design Guide (10/1/2025).
- SDS for ASTM A1003 Steel Framing Products for interior framing, exterior framing, and clips/accessories.
- The technical content of this literature is effective 08/26/25 and supersedes all previous information.

DISCLAIMER

All data, details, and specifications included herein are intended as a general guide for using Refined Metal products.
These products should not be used in design or construction without evaluation by a qualified engineer or architect to determine suitability for a specific use.
Refined Metal assumes no liability for failure resulting from use or misapplication of computations, details, or specifications contained herein.
Refined Metal assumes no liability for damages resulting from improper application or installation of these products.

DERIVED SMART STUD SPAN TABLES

Preliminary single-span estimates in feet. Verify with manufacturer span tables or sealed engineering before use.

	15 psf dead load, 40 psf live load		15 psf dead load, 60 psf live load		15 psf dead load, 125 psf live load	
Spacing OC	**L/360**	**L/480**	**L/360**	**L/480**	**L/360**	**L/480**
12	7.9	7.7	6.7	6.7	4.9	4.9
16	6.8	6.8	5.8	5.8	4.3	4.3
24	5.6	5.6	4.8	4.8	3.5	3.5

	40 psf dead load, 40 psf live load		40 psf dead load, 60 psf live load		40 psf dead load, 125 psf live load	
Spacing OC	**L/360**	**L/480**	**L/360**	**L/480**	**L/360**	**L/480**
12	6.5	6.5	5.8	5.8	4.5	4.5
16	5.7	5.7	5.1	5.1	3.9	3.9
24	4.6	4.6	4.1	4.1	3.2	3.2

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